

DEMENTIA

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Updated in September 2024 by Dr Hou Minsheng. Next review date: September 2027.

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INTRODUCTION

Importance

Dementia is a neurodegenerative disease that is characterised by progressive impairment of cognitive function. As the disease increases in severity, patients may experience some or all of the following: memory loss, language impairment, disorientation, changes in personality, difficulty with activities of daily living, self-neglect, neuropsychiatric symptoms and out of character behaviour.

Right from the stage of mild dementia up to the stages of severe dementia, this condition poses a significant health and socio-economic burden to patients, caregivers, and the nation as a whole. Early diagnosis will allow appropriate use of pharmacological and non-pharmacological management. Along with disease stabilization, efforts to address caregiver burden, patient safety and medico legal concerns should form the principles of management.

There are various causes of dementia which can be classified into irreversible and potentially reversible causes:

- a) Irreversible causes include degenerative causes (Alzheimer's disease, fronto-temporal dementia and dementia with Lewy body), cerebrovascular disease (Vascular dementia), dementia resulting from Parkinson's disease and other disorders such as Creutzfeld-Jakob disease. Alzheimer's disease (AD) and vascular dementia (VaD) are the most common causes of dementia.
- b) Potentially reversible causes include infectious disorders (meningitis and encephalitis), toxic or metabolic encephalopathies (hypothyroidism, vitamin B12 deficiency, and alcohol-related syndromes), neoplastic causes and hydrocephalus (obstructive or normal pressure hydrocephalus).

Epidemiology

Singapore has one of the fastest ageing populations in the Asia-Pacific region. By 2030, 15-20% of our total population will be at least aged 65. Dementia represents a major public health concern in Singapore and worldwide. Being the most prevalent neurodegenerative disease, dementia is expected to increase from 22,000 in 2005, to almost 53,000 in 2020, and 241,000 in 2050.

In Southeast Asia, the number of people with dementia is predicted to increase from 2.48 million in 2010 to 5.3 million in 2030. **Whereas worldwide**, it is estimated that there were 35.6 million people with dementia in 2010, with the numbers doubling every 20 years to 65.7 million in 2030 and 115.4 million in 2050.

The commonest cause of dementia is Alzheimer's disease with vascular dementia being the next most important cause. A number of cohort studies and systematic reviews have demonstrated that vascular risk factors contribute to an increased risk of not just vascular dementia but also Alzheimer dementia. In particular, mid-life hypertension (R.R 1.24-2.8), mid-life hypercholesterolaemia (R.R. 1.4-3.1), diabetes mellitus (R.R. 1.39-1.47) and strokes have all been shown to be associated with an increased risk of incident dementia. These risk factors represent possible windows of opportunity for intervention that may allow a reduction of dementia risk in the years ahead.

SCREENING AND DIAGNOSIS

Screening for Dementia in Teamlet / General Pool

Screening should be targeted at individuals with:

1. Complaints of memory or cognitive impairment. This could be either self-reported or reported by the caregiver or family members.
2. Suspicion of cognitive impairment by healthcare professionals during interaction despite the absence of memory complaints.
3. History of stroke or known risk factors of stroke.
4. Increased risk of dementia (such as those with a strong family history of dementia).
5. Questionable mental competency and who need to make important decisions (such as the sale of a house or making a will).
6. Assessment to review aspects such as fitness to drive (in the elderly driver).

Symptoms

The symptoms of cognitive impairment include:

- Progressive forgetfulness, especially short-term memory of information and events e.g. appointments, misplacing items.
- New problems with communication, e.g. repeating themselves in conversations, expressive aphasia
- Misplacing things and losing the ability to retrace steps, sometimes accusing others of stealing their belongings, or getting lost
- Confusion with time and place
- Difficulties performing daily activities, especially iADL
- Problems with planning or solving problems
- Impaired judgment, e.g. handling money
- Changes in mood and personality, e.g. suspicious of others
- Withdrawal from work or social activities
- Difficulty understanding visual images and spatial relationship, e.g. balance, reading, driving

Assessment in Teamlet / General Pool

1. Exclude delirium if the patient presents with forgetfulness or confusion of an acute nature.
2. Consider potentially reversible neurological conditions and depression if this is of a sub-acute nature (weeks to few months).
3. **Abbreviated Mental Test (AMT)**
 - a) The AMT is a 10-item screening test that has been validated locally to screen for dementia.
 - b) A score of 7 or less may suggest cognitive impairment in patients with primary school education or below, and should be further assessed.
 - c) Patients with secondary education or higher should be allowed one less error. A score of 8 or less needs to be assessed.

Items	Score
What is the year?	1
What is the time? (within 1 hour)	1
What is your age?	1
What is your date of birth?	1
What is your home address?	1
Where are we now?	1
Who is our country's Prime Minister	1
What is his / her job?	1
Memory phrase "37 Bukit Timah Road"	-
Count backwards from 20 to 1	1
Recall memory phrase	1
Total Score	

4. Dementia Work-up Panel

- Full blood count
- Vitamin B12
- Sodium, potassium and creatinine
- Liver function test
- Thyroid function test
- Corrected Calcium
- ECG: to exclude conduction problems which is a contraindication to Acetylcholineesterase inhibitor (AChEI) therapy
- Routine testing for neurosyphilis is not recommended and it is best done when patients exhibit clinical features of neurosyphilis or patients with history of sexually transmitted diseases

Basic investigations such as AMT and Dementia Work-up Panel should be done at clinic level. If patient is above 65 years old with more than 6-months of memory loss, refer him/her to Memory Clinic by ordering a paired Memory Clinic Nurse consult for assessments to be done prior to Doctor consult.

Criteria for Diagnosis of Dementia According to DSM-IV-TR

1. Memory impairment (amnesia) plus one of the following:
 - Aphasia
 - Apraxia
 - Agnosia
 - Disturbance in executive functioning
2. Cognitive impairments must be of sufficient severity to affect social and occupational functioning.
3. The cognitive deficits must represent a decline from a previously higher level of functioning.
4. The course is characterized by gradual onset and continuing cognitive decline.
5. The diagnosis of dementia should not be made if the cognitive deficits occur exclusively during the course of a delirium.

Criteria for Diagnosis of Dementia According to DSM-5

Major Neurocognitive Disorder (or Dementia)

- Evidence of **significant** cognitive decline from a previous level of performance in one or more cognitive domains (complex attention, executive function, learning and memory, language, perceptual-motor, or social cognition) based on:
 - Concern of the individual, a knowledgeable informant, or the clinician that there has been a **significant** decline in cognitive function; and
 - A **substantial** impairment in cognitive performance, preferably documented by standardized neuropsychological testing or, in its absence, another quantified clinical assessment
- The cognitive deficits **interfere** with independence in everyday activities (i.e. at a minimum, requiring assistance with complex instrumental activities of daily living such as paying bills or managing medications)
- The cognitive deficits do not occur exclusively in the context of a delirium
- The cognitive deficits are not better explained by another mental disorder (e.g. major depressive disorder, schizophrenia)

Mild Neurocognitive Disorder (or Mild Cognitive Impairment)

- As above, except that cognitive decline is **modest**, and cognitive deficits **do not interfere** with independence in everyday activities

Aetiology

Causes of dementia can be classified as irreversible and potentially reversible:

Irreversible	Potentially Reversible
Degenerative causes ○ Alzheimer's disease ○ Frontotemporal dementia ○ Dementia with Lewy bodies ○ Parkinson's Disease Dementia Cerebrovascular disease ○ Vascular dementia Prion-associated disorders Neurogenetic	Infectious disorders ○ Meningitis ○ Encephalitis Toxic or metabolic encephalopathies ○ Hypothyroidism ○ Vitamin B₁₂ deficiency ○ Alcohol-related syndromes Neoplastic causes Hydrocephalus

Initial Assessment of New Patients in Memory Clinic

Modified Chinese Mini-Mental State Examination (mcMMSE) by Nurse

mcMMSE is a cognitive test administered by care managers to assess patients for possible dementia. It is used as initial assessment for first visit to Memory Clinic and for re-assessment (when needed) during review visits, or during yearly review post discharge from Memory Clinic to track cognitive function. Memory Clinic Doctors will take reference from mcMMSE scores to aid in diagnosis of and assess severity of dementia. It may also be administered during functional assessments for patients with known dementia.

The score is calculated by assigning scores to the response categories of "Orientation", "Registration", "Attention and Calculation", "Recall", "Language", and "Praxis" respectively. Each "Yes" scores 1 and the total scores for 30 questions is at **30**. Cut-off depends on educational level.

Educational Level	Abnormal if Less Than
No Formal Education	20
Primary	22
Secondary / Tertiary	24

Severity of Dementia

Mild: 18 - 24, Moderate: 10 - 17, Severe: < 10

History taking by Memory Clinic Doctor

1. As far as possible, obtain information from caregiver or family members to verify patient's history.
2. Ask for:
 - Onset of cognitive decline –
 - Insidious (suggestive of Alzheimer's Disease).
 - Sudden (suggestive of vascular dementia).
 - Alcohol consumption if significant.
 - Use of medications anti-psychotics, antidepressants, anti-cholinergic agents, sedative-hypnotic agents.
 - History of medical, neurological and psychiatric illness.

Physical Examination and Mental State Examination by Memory Clinic Doctor

This includes looking for evidence of:

- Depression
- Focal neurological deficits (such as visual field defects, hemiparesis, hemisensory loss, asymmetric deep tendon reflexes or unilateral extensor plantar responses)
- Extrapyrimal signs (such as rigidity, bradykinesia, movement disorders and gait abnormalities)

A brief assessment may be conducted on **cognition using mcMMSE**, behavioural and functional problems, **and social and caregiver needs**.

When to do CT Scan for Patients with Memory Loss?

Canadian Consensus Conference on the Assessment of Dementia (CCCAD)

- Age less than 60 years
- Rapid (e.g. 1 or 2 months) unexplained decline in cognition or function
- “Short” duration of dementia (less than 2 years)
- Recent and significant head trauma
- Unexplained neurological symptoms (e.g. new onset of severe headache or seizures)
- History of cancer (especially in sites and types that metastasize to the brain)
- Use of anticoagulants or history of bleeding disorder
- History of urinary incontinence and gait disorder early in the course of dementia (as may be found in normal pressure hydrocephalus)
- Any new localizing sign (e.g. hemiparesis or a Babinski reflex)
- Unusual or atypical cognitive symptoms or presentation (e.g. progressive aphasia)
- Gait disturbance

Dementia Severity by Function

Mild	Need assistance in instrumental activities of daily living (e.g. managing money, marketing, housework, cooking)
Moderate Dementia	Need assistance in basic activities of daily living (e.g. feeding, toileting, bathing, dressing)
Severe Dementia	ADL dependent

MANAGEMENT

Initial Management

- Assessment using mcMMSE and consult to consolidate history from patients and caregivers. Refer to CT Scan if needed.
- If patient is diagnosed with Dementia, offer second assessment for completion of functional screening using Modified Barthel Index (MBI) and Lawton, Zarit Burden Interview Scale for Caregivers and identify the severity of the condition.

Recommended Follow-Up Interval for Re-assessment or Yearly Assessment

1. As dementia is a progressive deteriorating condition, at each visit, consider looking for complications in the following domains:

(i) **Affect and mood**

Anxiety and **depression** are common in the early stages. The doctor should routinely ask patient or caregivers for symptoms suggestive of depression, anxiety, paranoia or hallucinations.

(ii) **Behavioural and Psychological Symptoms in Dementia (BPSD)**

Behavioural problems occur frequently but tend to occur in the late stages. Specific behaviour to be aware of are wandering, verbal and non-verbal abuse, agitation, screaming and sleep problems. Most of the major behavioural problems are treated through non-pharmacological means using the ABC approach (**A** – Antecedent, **B** – Behaviour and **C** – Consequence) and later pharmacological methods as either an adjunct or second line therapy.

(iii) **Cognition**

The doctor should routinely ask patient or caregivers for any deterioration in memory, orientation, communication, recognition of people and objects, doing daily activities, judgment and problem solving activities. Consider repeating the **Modified Chinese Mini-Mental State Examination (mcMMSE)** by the care manager to look for any deterioration in MMSE score.

(iv) **Drugs**

Due to the age of the patients with dementia and their multiple co-morbidities, **polypharmacy** is common. The onset of new symptoms or worsening of existing symptoms may be due to adverse effects of drugs or its interaction with coexisting drugs and should not be just attributed to the worsening of the condition. Consider medication reconciliation to avoid polypharmacy where appropriate.

Assess for whether patient is on anti-cholinergic medications as these medications should be avoided, or at most, used sparingly in the elderly or in patients at risk of dementia). Examples of anti-cholinergic medications:

- Antidepressants e.g. amitriptyline, imipramine
- Antiemetics (commonly over-used for chronic giddiness) e.g. prochlorperazine
- Antimuscarinic (useful for overactive bladder) e.g. oxybutynin
- Antihistamines e.g. diphenhydramine, chlorpheniramine (piriton)

- Antiparkinsonian e.g. benztropine
- Antipsychotics e.g. olanzapine, quetiapine

(v) **Social environment**

Social issues may also result in coping difficulties by patients and caregivers. Consider routine screening for:

- Caregiver stress
- Elder abuse
- Financial difficulties

(vi) **Functional assessment**

When there are concerns with regards to home safety, driving safety, reports of recurrent falls, functional decline, swallowing difficulties, constipation, incontinence, malnutrition, *etc.*

Functional difficulties can be assessed at three levels: self-care, home functioning and community functioning. Refer to care manager for functional ability assessment if needed.

In view of the high risk of falls in these patients, all patients undergoing a functional assessment will be assessed for fall risk as well.

The interval of **re-assessment** for the above can range from 3 months to 6 months.

2. Consider referring to CM for **yearly assessment** for the following **at least once a year**:

- (i) **mcMMSE**
- (ii) **Modified Barthel Index**
- (iii) **Lawton-Brody Instrumental Activities of Daily Living Scale**
- (iv) **Zarit Burden Interview Scale (for main caregiver)**

Non-Pharmacological Management Principles of Patients Diagnosed with Dementia

Multi-component cognitive interventions along with physical exercise and training in activities of daily living are beneficial to patients living with dementia. Management strategies include:

1. Designing and maintaining a daily routine
2. Encouraging activities to engage the patient such as daily chores, creative or intellectual activities, physical activities.
3. Caregiver education and training should be considered to support caregivers in caring for the patients in the community.
4. Appropriate utilisation of community resources such as dementia day care centres, caregiver support groups, etc should be considered. If patients and caregivers are keen, consider referral to MSW for application for these services.

Pharmacological Management

1. Treating identifiable reversible causes
 - Treat depression (pseudo dementia)
 - Replace deficiency states (e.g. B₁₂ deficiency, hypothyroidism)
 - Correct metabolic abnormalities (e.g. hypocalcaemia, hypoglycemia)
 - Remove anti-cholinergic medications.
2. Reduction of vascular risk factors
 - Hyperlipidemia, hypertension, diabetes mellitus, smoking cessation, obesity
 - Anti-platelet agents for secondary stroke prevention
 - Anti-coagulation for atrial fibrillation and cardio embolic strokes
3. Slowing the rate of disease progression (Cognitive enhancers such as Acetylcholinesterase inhibitors (AChEI), e.g. Donepezil, rivastigmine and NMDA receptor antagonist, e.g. Memantine)
 - These should only be considered after a detailed discussion with the caregiver and patient (if possible), the risks and benefits of symptomatic treatment and the specific needs of the patient including cost considerations.
4. Complications arising from disease progression

Cognitive Enhancers are initiated at the hospitals or at NUP Memory clinics. Doctors are able to titrate the medication in the event of adverse events such as diarrhea, vomiting, nausea, giddiness, bradycardia or muscle weakness arising from the use of the medication.

Doctors may start antidepressant, anxiolytics, mood stabilizers and antipsychotics after discussion with the caregiver the benefits and the risk of such treatment.

All patients should be reviewed within **12** weeks upon initiation of new medication to assess the benefits and adverse effects of the medication. Patient and caregivers are also advised to seek early medical treatment if they develop adverse effects in the interim.

Examples of common cognitive enhancers used in Dementia

- Acetylcholinesterase inhibitors (AChEI) – donepezil, rivastigmine
- NMDA antagonists – memantine

Drug	Initial Dose and Titration	Maximum Dose	Common ADR	Contraindications / Precautions / Remarks
Acetylcholinesterase Inhibitors				
Donepezil (S2) \$\$^	Initial Dose: 5 mg/day	23 mg/day	Diarrhoea, nausea, vomiting, headache, anorexia, abnormal	Common 1st line medication for mild-moderate dementia (Alzheimer's/mixed)

Drug	Initial Dose and Titration	Maximum Dose	Common ADR	Contraindications / Precautions / Remarks
Strength per tablet: 5 mg 10 mg	Usual Dose: 5 - 10 mg/day Titration: 5 mg; 4 week interval		dreams, bradycardia, syncope, dizziness	5 mg OM is the usual starting dose. For patients at risk of side effects e.g. elderly, donepezil may be started at 2.5 mg OM for 4 - 6 weeks, before increasing to 5 mg OM. Aim for standard doses of 5 - 10 mg (avoid sub-therapeutic doses unless there are tolerability issues) Contraindications: bradycardia or known cardiac conduction system disease Precautions: peptic ulcer disease, COPD and/or asthma, seizure disorder, urinary tract obstruction
Rivastigmine (NS) (Exelon) \$\$\$\$\$\$^ Patch strength: 4.6 mg/24 hours 9.5 mg/24 hours <i>13.3mg/24 hours patch not available in NUP formulary</i>	Initial Dose: 4.6 mg/24 hours Usual Dose: 9.5 mg/24 hours Titration: 9.5 mg; 4 week interval	13.3 mg/ 24 hours	Diarrhoea, nausea, vomiting, headache, anorexia, agitation, abnormal dreams, bradycardia, syncope, dizziness + Contact rash, pruritus (for patch - rotate sites)	Aim for standard doses of (Patch) 9.5 mg OD (avoid sub-therapeutic doses unless there are tolerability issues) Contraindications: bradycardia or known cardiac conduction system disease Precautions: peptic ulcer disease, COPD and/or asthma, seizure disorder, urinary tract obstruction
Rivastigmine (NS) (Exelon) \$\$\$\$\$\$^ Oral: Not available in NUP	Initial Dose: 1.5 mg BD Usual Dose: 6 mg BD Titration: 1.5 mg; 4 week interval	12 mg/day		
NMDA Antagonists				
Memantine (S2) \$\$-\$\$\$^ Strength per tablet: 10 mg	Initial Dose: 5 mg/day Maintenance Dose: 10 mg BD or 20 mg daily Titration: 5 mg; 2 week interval	20 mg/day	Headache, dizziness, agitation, constipation, confusion	5 mg OM for 2 weeks is the usual starting dose, then 5 mg BD for 2 weeks, then 10 mg OM / 5 mg ON for 2 weeks, then 10 mg BD thereafter. If patient develops side effects, this interval can be further prolonged. Aim for standard doses of 10 mg BD (avoid sub-therapeutic doses unless there are tolerability issues) *Note that renal dose adjustment is required. Precautions: severe hepatic impairment, seizure disorder

Examples of common medications used for BPSD:

Non-pharmacological options are first line for BPSD, however, if pharmacologic agents are required, first line are antidepressants

Drug	Initial Dose and Titration	Maximum Dose	Common ADR	Contraindications / Precautions / Remarks
Selective Serotonin Reuptake Inhibitor (SSRI) Antidepressants				
Fluvoxamine (S2) \$-\$-\$^ (Faverin) Strength per tablet: 50 mg	Initial Dose: 25 mg – 50mg Usual Dose: 50 – 100mg/day Titration: 25 – 50mg; 1 week interval	100 mg/day Combination Maximum: Fluvoxamine 50mg/day and Mirtazapine 15mg/day	Common side effects: Nausea, vomiting, diarrhoea, dry mouth, nervousness, headache, dizziness. Usually, it lasts the first 1-2 weeks. Uncommon side effects: Sexual dysfunction, tremors, akathisia, hyponatremia, bleeding, QT prolongation, Serotonin syndrome, i.e. myoclonus, hyperreflexia – especially when in combination with TCAs, MAOI etc. Avoid antidepressant monotherapy in patients with bipolar disorder.	• Sedating, useful to help sleep • Avoid antidepressants with anticholinergic activity: Amitriptyline, Imipramine • Contraindication: use of MAOI concurrently or within 14 days of discontinuing Fluvoxamine or MAOI
Escitalopram (NS) \$\$\$ - \$\$\$^ (Lexapro) Strength per tablet: 10 mg	Initial Dose: 5 mg/day Usual Dose: 5 - 10 mg/day Titration: 5mg; 4 week interval	10 mg/day	Common side effects: Nausea, vomiting, diarrhoea, dry mouth, nervousness, headache, dizziness. Usually, it lasts the first 1-2 weeks. Uncommon side effects: Sexual dysfunction, tremors, akathisia, hyponatremia, bleeding, QT prolongation, Serotonin syndrome, i.e. myoclonus, hyperreflexia – especially when in combination with TCAs, MAOI etc. Avoid antidepressant monotherapy in patients with bipolar disorder.	• Activating • Avoid antidepressants with anticholinergic activity: Amitriptyline, Imipramine • Contraindication: use of MAOI concurrently or within 14 days of discontinuing Escitalopram or MAOI

Drug	Initial Dose and Titration	Maximum Dose	Common ADR	Contraindications / Precautions / Remarks
Alpha-2-antagonist Antidepressants				
Mirtazapine (S2) \$-\$\$^ (Remeron) Strength per tablet: 15 mg	Initial Dose: 15 mg Usual Dose: 15 – 30mg/day Titration: 7.5 – 15mg; 1 – 2 weeks interval	30 mg/day Combination Maximum: Fluvoxamine 50mg/day and Mirtazapine 15mg/day	Dry mouth, constipation, Sedation (7.5 - 15 mg more sedating than 30 mg), increase appetite, orthostatic hypotension, headache Avoid antidepressant monotherapy in patients with bipolar disorder.	• Sedating, improves appetite • Useful for patients with poor appetite, may cause weight gain • Avoid antidepressants with anticholinergic activity: Amitriptyline, Imipramine • Contraindication: use of MAOI concurrently or within 14 days of stopping an MAOI • Blood dyscrasia (check Dementia panel result before starting) • Less interaction with warfarin as compared to Fluvoxamine
Sedatives				
Zopiclone (NS) \$^ Strength per tablet: 7.5 mg	Usual Dose: 3.75mg ON/PRN for sleep	7.5mg ON PRN	Sedation, nausea, vomiting, dry mouth, dizziness, headache Use can lead to addiction, abuse, and misuse	• Short course, not more than 2 weeks • Contraindication: severe respiratory impairment, myasthenia gravis, severe hepatic insufficiency, history of complex sleep behaviours after taking nonbenzodiazepine sedative/ hypnotic
Antipsychotics				
Quetiapine (S2) \$ - \$\$\$^ (Seroquel) Strength per tablet: 25 mg 100 mg	Initial Dose: 12.5mg – 25mg/day Usual Dose: 12.5–25 mg/day Titration: 6.25 – 12.5mg; 1 week interval	75mg BD	Sedation, nausea, vomiting, constipation, dry mouth, orthostatic hypotension, headache, weight gain	There is an FDA black box warning for antipsychotics and adverse cardiovascular events. As such, use beyond 12 weeks is not recommended Preferred atypical antipsychotic in patients at high risk of extrapyramidal symptoms

^ Cost

\$ <\$10 per month
 \$\$\$\$ \$30 - <\$60 per month

\$\$ \$10 - <\$20 per month
 \$\$\$\$\$ \$60 - <\$90 per month

\$\$\$ \$20 - <\$30 per month
 \$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

Role of Health Team Members

1. Family Physician

- Able to provide all aspects of primary medical care from screening, diagnosis and management of dementia, including health promotion and prevention / treatment of complications
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care

2. Care Manager

- Provide basic education and materials on dementia for patients and caregivers
- Dementia assessment: mcMMSE, modified Barthel Index (ADLs), Lawton Instrumental Activity of Daily Living (iADLs)
- Screen for caregiver burden via the Zarit Burden Interview Scale
- Provide functional assessment when necessary

3. Advanced Practice Nurse

- Co-manage patients who are diagnosed with mild cognitive impairment (MCI) and stable dementia, including pre-existing chronic conditions with collaborating physicians and health team
- Liaise with other healthcare personnel to close care gaps and provide holistic and patient-centred care

4. Medical Social Worker

- Conduct biopsychosocial assessment
- Provide intervention for social assistance including application of schemes and services, and referral for community resource and caregiver support
- Consider alternative financial support (not related to medical bills) when necessary
- Referrals to Dementia related care in the community when needed, e.g. Day Care Centre and Nursing Home

5. Pharmacist

- Monitoring of Memory Clinic patients who have started on new medications and/or have medication dose changes
- Provide patient education and counselling
- Smoking cessation clinic
- Detect, prevent and address drug-drug/drug-disease interactions
- Perform medication reconciliation.

6. Care Coordinator

- Introduce OneNUHS and HealthHub Apps to patient
- Introduce and enrol patient onto Teleconsult
- Perform PHQ2, frailty and functional screenings, abbreviated mental test (AMT) and comprehensive geriatric assessment (CGA) when indicated

- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza / pneumococcal), cancer screenings (FIT kit, mammogram, PAP smear / HPV), DRP and DFS

7. Financial Counsellor

- Financial counselling for patients with issues on payment of medical bills
- Advised on MediSave utilisation for chronic disease medical bills
- Application on government endowment fund- MediFund and Medication Assistance Fund for subsidies on high cost drugs
- Application on Institutional Medical Fund-Remission for eligible non-residents

8. Psychologist

- Psychological assessment and intervention to manage mood challenges (depression, anxiety etc.) and behavioural issues associated with dementia
- Support patient and caregiver in developing strategies to improve daily functioning and quality of life
- Psychological support for carers experiencing caregiver stress, depression and anxiety

9. Dietitian

- Dietary management for patients with malnutrition or poor appetite / oral intake

10. Physiotherapist

- Strengthening exercises as falls are common in elderly patients with dementia

SPECIAL SITUATIONS

Referral to NUP Memory Clinic

- **Inclusion criteria**
 - Above 65 years old with memory problems
 - Memory loss > 6/12
- **Exclusion criteria**
 - Legal issues / LPA / requires neuropsychological testing or mental capacity assessment (refer Psychiatry)
 - < 65 years old, refer Neurology for early onset dementia
- **Offer Memory Clinic assessment**
 - Ensure Dementia panel is done and normal
 - Order TCU NUR CM Memory Clinic and TCU Dr Memory Clinic (1st visit)

- Advise to come with caregiver
- **Dementia Assessment done by CM when patient is referred to Memory Clinic**
 - mcMMSE
 - Modified Barthel Index (ADLs)
 - Lawton Instrumental Activity of Daily Living (iADLs)
 - Zarit (for caregiver)

Discharge Criteria from Memory clinic to General Pool (or teamlet, if applicable)

- Diagnosis made
- BPSD well managed
- Nil medication issues or side effects to medications
- Caregiver stress addressed
- Dementia assessment and Zarit score completed in the past 1 year
- Family and patient agreeable
- As memory and functional decline will be expected, yearly TCU NUR CM Consult (ordered by general pool doctors after discharge from Memory clinic) for dementia assessment is still recommended in the monitoring of dementia patients and their caregivers

Referral Back to NUP Memory Clinic

- Sudden drastic decline in memory, patient otherwise well (or MMSE dropped > 4 points per year; usual expected decline is 1-2 points per year)
- BPSD surfaces or worsens
- Caregiver stress and burn-out
- If more social support is needed, Memory clinic team can work directly with NUHS G-RACE Manager to link up with community partners to provide home visits and caregiver support at the patients' homes

Special Precautions in Dementia Patients with Chronic Diseases

Patients with dementia should be managed holistically for their other chronic diseases, if any. Special care should be exercised in pharmacological management, as dementia and chronic disease medication may interact in many ways.

1. Diabetes

- Do not aim for excessively tight glycemic control in frail elderly patients with dementia who can be at risk of developing hypoglycemia.
- Caregivers should help with insulin injection for patients who require insulin.

2. Hypertension / Cardiac Arrhythmia

- In patients on AChEI, Calcium channel blockers or beta blockers can cause or worsen bradycardia and should be avoided in patients with known heart block or other cardiac arrhythmias.

3. COPD / Asthma

- AChEI can cause bronchoconstriction. Close monitoring of lung condition is needed for patients with dementia started on AChEI if the patients also have concomitant asthma / COPD.

4. Parkinson's Disease

- Typical antipsychotics have extrapyramidal side effects and can worsen the symptoms of Parkinson's disease.

5. Renal Impairment

- Memantine should be avoided if Cr > 200 $\mu\text{mol/L}$ or eGFR < 30 ml/min. Maximal dose should be limited to 5mg BD for those with eGFR 30 - 60ml/min.

Medical Report

For application to community services such as Nursing Home, Medical Social Worker may approach doctor to provide information such as diagnosis, clinical history, required nursing procedures and X-Ray results in patient's Medical Report.

REFERRALS AND COMMUNITY RESOURCES

Referral to EMD

- Less than 3 months duration and associated with sudden onset of neurological deficits e.g. unilateral weakness, slurring speech, imbalance, fall, incontinence, disorientation, confusion OR fluctuating memory (sometimes remember, sometimes not)
- When delirium is suspected
- When patient is causing significant harm to self or others

Referral to Specialists

- Complicated cases of dementia, e.g. Severe BPSD
- Patients who require specialist opinion for medication titration for their condition (e.g. side effects or complications from conventional medication, requiring long-term sleeping pills, on optimal BPSD medications (e.g. on both Fluvoxamine 50mg and Mirtazapine 15mg) but still symptomatic)
- Development of geriatric problems that require a higher level of care, e.g. severe constipation, fecal impaction, frequent micturition, nocturia, etc.
- Legal issues / LPA / mental capacity assessment (special arrangement with NUH Psy)
- **NUH Memory Clinic (Psycho-Geriatric):**
 - PSYCHIATRIC symptoms prominent e.g. depression, hallucination, delusion, paranoia against domestic helper and relatives, aggressiveness, sleep reversal
 - Case managers from NUH G-RACE may be engaged to arrange social and psychological support when there is no suitable community partner
- **NUH Memory Clinic (Geriatric):**
 - GERIATRIC symptoms prominent e.g. incontinence, fall, frailty, poor feeding

Referral to Community Partners

- Eldercare services and caregiver support such as meal services, home help, home visit, befriender services, case management, home medical, home nursing and social day care services. Example Fei Yue, NTUC Healthcare, Singapore Anglican Community Services (SACS).
- Dementia day care centres and nursing homes

RECOMMENDED CARE COMPONENTS

The domains of follow up care are summarised in the CDMP (Chronic Disease Management Program) table below:

Recommended Care Components	Minimum Frequency	Remarks
Assessment of Memory	Annually	For patients on cognitive enhancers, objective documentation of memory assessment with a bedside cognitive screening instrument (e.g. MMSE) must be performed.
Assessment of Mood and Behaviour	Annually	Enquiring about mood and behaviour and initiating appropriate non-pharmacological and/or pharmacological treatment where appropriate.
Assessment of Social Difficulties and Caregiver stress (if any)	Annually	Assessment and referral to care coordinator, medical social worker or appropriate community services may be required.
Functional Needs Assessment	Annually	To assess home safety, driving safety, falls, functional decline and swallowing difficulties.
Influenza Vaccination	Annually or per season	As recommended under the National Adult Immunisation Schedule (NAIS).

ABBREVIATIONS

AChEI	Acetylcholinesterase inhibitors
ADL	Activity of Daily Living
AMT	Abbreviated Mental Test
BPSD	Behaviour and Psychological Symptoms in Dementia
ECG	Electrocardiogram
Lawton	Lawton-Brody Instrumental Activities of Daily Living Scale
MBI	Modified Barthel Index
mcMMSE	Modified Chinese Mini-Mental State Examination
Zarit	Zarit Burden Interview Scale

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ACKNOWLEDGEMENTS

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